

COMPUTER ORGANIZATION AND ARCHITECTURE

GROUP NO : 5

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ABSTRACT

- Cache memory improves processor performance by reducing memory access latency.
- It acts as a high-speed buffer between the CPU and main memory.
- Cache organization significantly influences instruction and data access efficiency.
- Two major architectures:
 - Unified Cache
 - Split Cache (Instruction Cache & Data Cache)
- Unified cache stores both instructions and data in a single cache.
- Split cache separates instruction and data caches, allowing parallel access.
- Performance comparison is analyzed using the Ripes RISC-V simulator.

INTRODUCTION

- Modern CPUs operate much faster than main memory.
- This speed difference creates a memory bottleneck.
- Cache memory is used to reduce Average Memory Access Time (AMAT).
- Cache works based on Locality of Reference:
 - Temporal locality
 - Spatial locality
- Cache organization affects execution efficiency and system performance.
- This project studies the performance impact of unified and split cache designs.

RIPES SOFTWARE

- Ripes is a graphical simulator for RISC-V processor architecture.
- Used for learning computer organization and architecture concepts.
- Allows execution of RISC-V assembly programs.
- Provides visualization of:
 - CPU datapath
 - Pipeline stages
 - Cache behavior
- Supports simulation of different cache configurations.
- Used in this project to compare Unified Cache and Split Cache performance.

Program 1: Unified Cache Study

```
.data
array: .word 5,10,15,20,25,30,35,40
n: .word 8
```

```
.text
.globl main
```

```
main:
    la t0, array
    la t5, n
    lw t1, 0(t5)

    li t2, 0
    li t3, 0
```

```
loop:
    beq t2, t1, exit
    lw t4, 0(t0)
    add t3, t3, t4
    sw t3, 0(t0)
    addi t0, t0, 4
    addi t2, t2, 1
    j loop

exit:
    nop
```

Program for Split Cache Study

```
.data
value: .word 5

.text
.globl main

main:
    la t0, value
    lw t1, 0(t0)

    li t2, 100
```

```
loop:
    add t1, t1, t1
    addi t1, t1, 3
    addi t1, t1, 2
    addi t1, t1, 1
    sw t1, 0(t0)

    addi t2, t2, -1
    bnez t2, loop

exit:
    nop
```

RESULT

Unified Cache

- Single cache stores both instructions and data
- Combined memory accesses are handled by one cache
- Cache statistics observed in the simulator:

Metric	Values
Hits	16
Misses	1
Hit Rate	94.12%

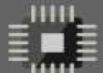
FileEditViewHelp



100 ms



100
1010
01
Editor

Processor

Cache

Memory

I/O

L1 Data CacheL1 Instr. Cache

Cache configuration:

Preset:



2^N Lines:

4

Repl. policy:

LRU

Ways:

1

Wr. hit:

Write-back

2^N Words/Line:

6

Wr. miss:

Write allocate

Plot configuration:

Numerator

Hits

Denominator

Access count

☒ Ratio

☒ Moving avg.

50 cyc.

Statistics:

Size (bits):

33136



Hit rate:

.9412

Writebacks:

0

Hits:

16

Misses:

1

TotalMoving average

Split Cache

- Separate instruction cache and data cache
- Cache statistics observed in the simulator:

Cache	Hits	Misses	Hit Rate
Instruction Cache	907	1	99.89%
Data Cache	100	1	99.01%

Instruction Cache :

File Edit View Help

100 ms

100
1010
01
Editor

Processor

Cache

Memory

I/O

L1 Data Cache L1 Instr. Cache

Cache configuration:

Preset:  

2^N Lines: Repl. policy:

Ways: Wr. hit:

2^N Words/Line: Wr. miss:

Plot configuration:

Numerator

Denominator

☒ Ratio

☒ Moving avg.

Statistics:

Size (bits): 

Hit rate: Writebacks:

Hits: Misses:

 Total  Moving average

Data Cache:

File Edit View Help

100 ms

100 1010 01 Editor

Processor

Cache

Memory

I/O

L1 Data Cache L1 Instr. Cache

Cache configuration:

Preset: [dropdown] [save] [close]

2^N Lines: [2] Repl. policy: LRU

Ways: [1] Wr. hit: Write-back

2^N Words/Line: [6] Wr. miss: Write allocate

Plot configuration:

Numerator: Hits

Denominator: Access count

☒ Ratio

☒ Moving avg. 50 cyc.

Statistics:

Size (bits): 8292 [help]

Hit rate: .9901 Writebacks: 0

Hits: 100 Misses: 1

■ Total ■ Moving average

Result Analysis :

- In Unified Cache, instructions and data compete for the same cache space.
- This increases the probability of conflict misses.
- Instruction fetch and data access cannot occur simultaneously.
- In Split Cache, instructions and data are stored in separate caches.
- This enables parallel instruction fetch and data access.
- As a result, split cache shows better execution efficiency and reduced cache conflicts.

CONCLUSION

- Cache organization has a major impact on processor performance.
- Unified cache
 - Simpler architecture
 - Lower hardware complexity
 - Possible instruction–data conflicts
- Split cache
 - Separate instruction and data access paths
 - Reduced conflicts and better efficiency
- Modern processors typically use split cache architecture for higher performance.

THANK YOU